# STA301 Statistics and Probability

## Final Term Examination - August 2004

**Time Allowed: 150 Minutes**

## Instructions

**Please read the following instructions carefully before attempting any of the questions:**

1. Attempt all questions. Marks are written adjacent to each question.
2. Do not ask any questions about the contents of this examination from anyone.
   a. If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.
   b. If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.
   c. Write all steps, missing steps may lead to deduction of marks.
3. You are allowed to use the calculator & Statistical tables in order to solve the questions.
4. For your convenience we are providing you the following symbols,

□ ,?, $\bar{X}$ or write Mean, s, ∫ or sd for standard deviation, s2 ∫2 or sd2 or variance for variance, $\sqrt{\ }$ for square root or whole square root.

****WARNING: Please note that Virtual University takes serious note of unfair means. Anyone found involved in cheating will get an `F` grade in this course.**

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**Total Marks: 50** **Total Questions: 15**

**Question No. 1** **Marks : 3**

Write down any three types of average.

**Question No. 2** **Marks : 3**

Define the Trial and the Outcome.

**Question No. 3** **Marks : 3**

What is meant by Sampling Distribution?

**Question No. 4** **Marks : 3**

Define the following:

1. Null Hypothesis
2. Alternative Hypothesis

**Question No. 5** **Marks : 1**

Which of the following is the example of discrete variable?

- The number of children in different families of a locality
- Ages of cancer patients
- Height of students
- None of these

**Question No. 6** **Marks : 1**

The Mode of the letters in the word STATISTICS is:

- S
- T
- I
- S and T

**Question No. 7** **Marks : 1**

In Binomial Experiment, successive trails are:

- Both independent and dependent.
- Dependent
- Independent
- None of above

**Question No. 8** **Marks : 1**

In a Binomial Distribution n = 10, p = 3/5. its mean will be:

- o 12/5
- o 1
- o 2/5
- o 6

**Question No. 9** **Marks : 1**

A population that consists of unlimited number of elements is called:

- o Finite population
- o Infinite population
- o Hypothetical population
- o Target population

**Question No. 10** **Marks : 1**

An estimator which is free from Bias is called:

- o Biased Estimator
- o Unbiased Estimator
- o Consistent Estimator
- o Efficient Estimator

**Question No. 11** **Marks : 1**

The probability of accepting the true null hypothesis is called:

- o Level of significance
- o Level of confidence
- o Type-I Error
- o Type-II Error

**Question No. 12** **Marks : 1**

If the population standard deviation is not known and sample size is large (n > 30), then the test statistic to be used is:

- o t-test
- o Z-test
- o F-test
- o $\chi^2$-test

**Question No. 13** **Marks : 10**

Calculate the Mean Deviation from mean for the following data.

| Size of items | 3-4 | 4-5 | 5-6 | 6-7 | 7-8 |
|---|---|---|---|---|---|
| Frequency | 3 | 7 | 22 | 60 | 8 |

**Question No. 14** **Marks : 10**

The probability that a patient recovers from a delicate heart operation is 0.9. What is the probability that exactly five of the next 7 patients having this operation survive.(Use Binomial Distribution)

**Question No. 15** **Marks : 10**

A standard examination has been given for several years with $\mu$ = 70 and $\sigma^2$ = 9.A school using this examination for the first time, gave it to a group of 25 students who obtained a mean $\bar{X} = 71$ and a variance of $S^2 = 12$. Test the hypothesis that $H_0 : \sigma^2 = 9$ against the alternative $H_1 : \sigma^2 \neq 9$ at 5% level of significance.